ModelIt!

Teacher's Guide

Newton's Laws: Why Things Move the Way They Do

The Three Rules That Govern Every Motion in the Universe

Quick Reference

Grade Level	Grade 07L2
NGSS Standard	MS-PS2-1, MS-PS2-2
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: MS-PS2-1, MS-PS2-2

Apply Newton's Third Law to design a solution to a problem involving the motion of two colliding objects. Plan an investigation to provide evidence that the change in an object's motion depends on the sum of the forces acting on the object and the mass of the object.

Three-Dimensional Learning

Science Practice	**Planning and Carrying Out Investigations** — Students plan investigations to provide evidence that an object's change in motion depends on the sum of forces and the mass of the object.
Disciplinary Core Idea	**PS2.A Forces and Motion** — The motion of an object is determined by the sum of the forces acting on it. If the total force is not zero, the object will change speed and/or direction.
Crosscutting Concept	**Stability and Change** — Students analyze how the balance of forces determines whether an object maintains its current motion or changes speed and direction.

CAST Testing Alignment

MULTIPLE CHOICE:

Two identical forces are applied to two different objects. Object A has a mass of 2 kg and Object B has a mass of 8 kg. Which statement correctly compares their accelerations and why?

CONSTRUCTED RESPONSE:

Using your force and motion model, explain why a loaded truck takes longer to stop than an empty car traveling at the same speed. Reference net force, mass, and friction in your explanation.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don’t need to teach all of this — it’s here so you can answer student questions accurately.

Newton's Laws of Motion are the foundation of classical mechanics, explaining why objects move the way they do. From a rolling ball to a launching rocket, these three laws govern every motion in the universe at everyday speeds.

Newton's First Law: Inertia

An object at rest stays at rest, and an object in motion stays in motion at constant velocity, unless acted upon by an unbalanced force. This property of matter is called inertia. Mass is the measure of inertia: more mass means more resistance to changes in motion. A bowling ball has more inertia than a tennis ball.

Newton's Second Law: F = ma

The acceleration of an object is directly proportional to the net force acting on it and inversely proportional to its mass. Doubling the force doubles the acceleration. Doubling the mass halves the acceleration. This simple equation (F = ma) predicts the motion of everything from atoms to planets.

Newton's Third Law: Action-Reaction

For every action force, there is an equal and opposite reaction force. When you push on a wall, the wall pushes back on you with equal force. When a rocket expels gas downward, the gas pushes the rocket upward. These paired forces act on DIFFERENT objects.

Friction: The Hidden Force

Friction is a force that opposes the relative motion of two surfaces in contact. It converts kinetic energy to thermal energy (heat). Without friction, you couldn't walk, drive, or pick up objects. The amount of friction depends on the surfaces involved and the normal force pushing them together.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**
Students identify applied force, object mass, acceleration, friction force, and net force as components of the motion system.

E **Establish Relationships**
Students determine that applied force increases net force (and thus acceleration) while friction and mass oppose it, creating a balance that determines motion.

V **Visualize & Model**
Students build a model showing how F = ma governs motion, with friction as a real-world complication that reduces net force.

E **Evaluate Outcomes**
Students run light object, heavy object, and high friction scenarios to verify the F = ma relationship through different variable combinations.

R **Revise & Extend**
Students add air resistance or momentum to explore how additional forces affect motion in more realistic scenarios.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"Every time you throw a ball, ride a bike, or sit in a chair, you're experiencing Newton's Laws."*

Watch For

Drop a heavy and light object simultaneously. Ask: Which hits first?

Slide 2: Learning Objectives

3 min

Say: *"Today we're discovering the three rules that explain every motion in the universe."*

Watch For

Have students read objectives. Pre-teach 'net force' and 'acceleration.'

Slide 3: Big Question

3 min

Say: *"Push an empty cart. Easy. Now load it up. Much harder. Same push, different result. Why?"*

Watch For

Demonstrate with classroom objects of different masses. Discuss observations.

Slide 4: LEVER Framework

2 min

Say: *"We'll build a model that shows how force, mass, and friction determine ALL motion."*

Watch For

Preview LEVER steps. Connect to everyday motion experiences.

Slide 5: Activity 1: Components

8 min

Say: *"What determines how fast something accelerates? What can we control vs. what responds?"*

Watch For

Guide sorting. Discuss why applied force and mass are the inputs.

Slide 6: Activity 2: Connections

8 min

Say: *"When applied force goes up, what happens to net force and acceleration? But what does friction do?"*

Watch For

Help students identify positive and negative relationships. Introduce F = ma.

Slide 7: Activity 3: Simulation

10 min

Say: *"Let's push a light object and a heavy object with the same force. Who predicts what happens?"*

Watch For

Students predict first, then run all three scenarios. Graph and compare.

Slide 8: Discoveries

5 min

Say: *"So F = ma isn't just a formula. It EXPLAINS why pushing a full cart is harder. What else does it explain?"*

Watch For

Lead discussion connecting model to real-world examples: cars, rockets, sports.

Slide 9: STEM Challenge

12 min

Say: *"You're safety engineers now. Design a crash test that proves Newton's Laws save lives."*

Watch For

Groups design and conduct scaled crash tests with toy cars.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Applied Force, Object Mass
- **INTERNAL:** Acceleration, Friction Force, Net Force

Relationships

- **Applied Force to Net Force:** POSITIVE (+) — Increasing the applied force directly increases the net force available to accelerate the object, assuming friction stays constant.
- **Friction Force to Net Force:** NEGATIVE (-) — Friction opposes motion and subtracts from the applied force. Higher friction means less net force is available to accelerate the object.
- **Net Force to Acceleration:** POSITIVE (+) — Newton's Second Law: acceleration is directly proportional to net force. More net force produces more acceleration (F = ma).
- **Object Mass to Acceleration:** NEGATIVE (-) — Newton's Second Law: acceleration is inversely proportional to mass. More mass means less acceleration for the same net force (a = F/m).
- **Acceleration to Friction Force:** POSITIVE (+) — As an object moves faster (higher acceleration leads to higher velocity), friction force increases because there is more relative motion between surfaces.

Key Discoveries

- Acceleration depends on BOTH force AND mass: more force means more acceleration, but more mass means less acceleration (F = ma)
- Friction always opposes motion, reducing net force and therefore reducing acceleration
- Net force is what determines motion: an object only accelerates when forces are UNBALANCED
- A heavier object has more inertia, meaning it resists changes in motion more than a lighter object

Multiple Choice Post-Assessment Answer Key

1. Two identical forces are applied to two different objects. Object A has a mass of 2 kg and Object B has a mass of 8 kg. Using the model's demonstration of F = ma, which statement correctly compares their accelerations?

Correct: B) Object A accelerates four times faster than Object B because acceleration is inversely proportional to mass

Correct! Newton's Second Law states a = F/m. With the same force, Object A (2 kg) accelerates at F/2, while Object B (8 kg) accelerates at F/8. Since F/2 is four times F/8, Object A accelerates four times faster. Acceleration is inversely proportional to mass.

2. In the model, increasing Friction Force causes Net Force to decrease even when Applied Force stays constant. A student observes that a hockey puck slides much farther on ice than a ball rolls on grass with the same initial push. Which model relationship explains this?

Correct: B) Friction Force has a negative relationship with Net Force, and ice produces much less friction than grass, so more net force remains to sustain motion

Correct! Friction opposes motion and subtracts from applied force. On ice, friction is minimal, so nearly all of the applied force remains as net force. On grass, high friction subtracts most of the applied force, leaving little net force to sustain motion. This is why the puck slides farther.

3. A loaded truck traveling at 60 mph takes much longer to stop than an empty car traveling at the same speed, even when both apply maximum brakes. Which combination of Newton's Laws best explains this observation?

Correct: B) Newton's First Law (greater mass means greater inertia resisting the change in motion) combined with Newton's Second Law (same braking force applied to greater mass produces less deceleration)

Correct! The truck has greater mass, which means greater inertia (First Law: it resists changing its motion more). When brakes apply force, the truck's larger mass means less deceleration for the same force (Second Law: a = F/m). Both laws work together to explain the longer stopping distance.

4. In the model, a student sets Applied Force to 10 N and Object Mass to 5 kg, but observes that the actual Acceleration is less than the expected 2 m/s squared. Which model component accounts for the difference between expected and observed acceleration?

Correct: B) Friction Force, which opposes motion and reduces the net force available for acceleration below the full applied force

Correct! The calculation F/m = 10/5 = 2 m/s squared assumes no friction. In reality, friction subtracts from the applied force before it can accelerate the object. If friction is 3 N, the net force is only 7 N, producing acceleration of 7/5 = 1.4 m/s squared. The model captures this real-world complication.

5. A student pushes a friend on a skateboard. Newton's Third Law states that the friend pushes back on the student with equal force. The student asks: 'If the forces are equal, why does anyone move?' Which response correctly resolves this apparent paradox?

Correct: B) The action and reaction forces act on different objects, so they cannot cancel each other out. Each object accelerates based on the net forces acting on IT

Correct! This is a common misconception about Newton's Third Law. The action-reaction forces act on DIFFERENT objects. The push on the friend accelerates the friend (based on the friend's mass). The reaction push on the student accelerates the student backward (based on the student's mass). Forces only cancel when they act on the same object.

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Heavier objects fall faster"	In the absence of air resistance, ALL objects fall at the same rate regardless of mass. Galileo demonstrated this over 400 years ago. Air resistance affects light, large-surface-area objects more, which is why a feather falls slowly. In a vacuum, a feather and a bowling ball hit the ground at the same time.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"An object needs constant force to keep moving"	Newton's First Law says an object in motion stays in motion WITHOUT any force, unless something (like friction) acts on it. On Earth, friction always acts, so we think things 'naturally' stop. But in space, objects drift forever without any force at all.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Action-reaction forces cancel out"	Newton's Third Law forces act on DIFFERENT objects, so they never cancel. When you push a wall, you push the wall forward and the wall pushes you backward. These forces are equal and opposite but act on different objects. Forces only cancel when they act on the SAME object.